llmXive Automated Review of Qwen-VLA: Unifying Vision-Language-Action Modeling across Tasks, Environments, and Robot Embodiments

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper's factual claims regarding numerical results are largely consistent with the provided tables. The abstract correctly reports values such as 97.9% on LIBERO, 73.7% on Simpler-WidowX, and 86.1/87.2% on RoboTwin, which align with Table 1 (e002). Similarly, the OOD success rates for ALOHA (76.9%), SimplerEnv-OOD (32.0%), and DOMINO (26.6%) match Tables 2, 3, and 4 respectively.

However, there are three specific issues regarding the accuracy of comparative claims and their attribution:

1. **Ranking Accuracy (Abstract & Section 5.1.1):** The abstract states the model "matches the best on LIBERO (97.9%)". Table 1 clearly lists ABot-M0 with a score of 98.6%. While 97.9% is a strong result, claiming it "matches the best" is factually incorrect as it is 0.7 percentage points lower than the highest reported score. The text should be revised to state it is "competitive with the best" or explicitly note it is the second-best result.
2. **Statistical Attribution (Section 5.1.2):** The text claims the model "outperforms pi_0.5 and GR00T" and immediately follows with a "+35.4pp" gain. The calculation 76.9% (Qwen-VLA) - 41.5% (pi_0.5) equals 35.4pp. However, the gain over GR00T (25.4%) is 51.5pp. The current phrasing risks implying the 35.4pp margin applies to both baselines. The sentence should explicitly link the statistic to the pi_0.5 comparison only.
3. **Contextual Clarity (Abstract):** The abstract aggregates "real-world ALOHA" OOD results (76.9%) without explicitly distinguishing them from the simulation-based OOD benchmarks (SimplerEnv-OOD, DOMINO) mentioned in the same sentence. While the values are distinct, clarifying that the 76.9% figure is specific to the real-world ALOHA evaluation would prevent readers from misinterpreting it as an aggregate across all OOD benchmarks.

No citations were found to be unsupported by the internal data, but the phrasing of the "best" claim and the attribution of performance margins require correction to ensure factual precision.

Action items.

- [**writing**] Abstract claims model 'matches the best' on LIBERO (97.9%), but Table 1 shows ABot-M0 at 98.6%. Clarify that it is second-best or competitive, not the best.
- [**writing**] Section 5.1.2 states the model outperforms pi_0.5 and GR00T, then cites a +35.4pp gain. This gain applies only to pi_0.5 (76.9% vs 41.5%), not GR00T (25.4%). Clarify the specific comparison for the statistic.
- [**writing**] Abstract cites '76.9% average OOD success in real-world ALOHA'. Ensure this is clearly distinguished from simulation OOD results (SimplerEnv, DOMINO) to avoid

aggregation confusion.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively communicates the high-level architecture of Qwen-VLA, clearly mapping the model components (Qwen3.5 VLM, Diffusion Transformer) to the input modalities (Observed Images, Prompt, Noisy Action) and output (Clean Action). The visual layout successfully illustrates the unified nature of the model across manipulation, navigation, and vision-language understanding tasks as described in the caption.

Figure 2

The figure effectively visualizes the four-stage training pipeline described in the caption, but it includes an unexplained architectural block on the left and lacks a legend to define the icons used to indicate module states.

Figure 3

The figure contains a factual error in the top row where the caption describes 'two green staplers' while the image shows blue and yellow ones. Additionally, the caption fails to describe the middle row task shown in the figure.

Figure 4

Figure 4 provides a clear visual overview of the tasks and generalization capabilities of the model, but the caption is overly brief and the legend could be better aligned for readability.

Figure 5

The figure is visually clear with appropriate axes and legends, but the caption is truncated mid-sentence and the line plot lacks error bars for the reported success rates.

Figure 6

Figure 6 effectively visualizes the ablation studies, but the caption for panel (c) is incomplete and the labels in panel (b) appear swapped relative to the textual description of the optimal configuration.

Figure 7

Figure 7 provides a visual demonstration of generalization but contains significant discrepancies between the caption's description of objects and tasks (e.g., 'blue umbrella', 'green broccoli', 'bottled yogurt') and the actual content of the rendered images.

Action items.

- [**writing**] Figure 2: The diagram contains a large, unexplained block on the far left labeled 'Qwen-VLA' with 'Qwen3.5 VLM' and 'DiT' components that is not referenced in the caption or the Stage I-IV flow, creating confusion about whether this represents the model architecture or an additional training phase.
- [**writing**] Figure 2: The legend defining the symbols (e.g., the flame icon for 'unfrozen' or 'active' modules) is missing from the figure itself and not defined in the caption, making it difficult to interpret the status of the VLM and DiT blocks in Stages II, III, and IV.
- [**science**] Figure 3: The top row caption describes the task as 'Place the two green staplers side by side,' but the visual evidence shows the robot manipulating a blue stapler and a yellow stapler, not two green ones. This contradicts the visual data shown.
- [**science**] Figure 3: The middle row depicts a task involving a 'cake server' (as per the image text), but the caption only describes the top and bottom rows, omitting the middle row entirely.
- [**writing**] Figure 4: The caption 'Task Overview' is too brief to describe the complex content shown, which includes six specific tasks, generalization categories, and visual examples.
- [**writing**] Figure 4: The legend at the top uses color swatches to map to tasks, but the text labels are not aligned with the swatches, making it slightly harder to read.
- [**science**] Figure 4: The figure mixes 'In-Domain Tasks' (center) with 'Generalization' examples (sides) without clearly distinguishing them as separate experimental conditions or categories.
- [**fatal**] Figure 5 caption is truncated mid-sentence at the end of part (b) ('...when pairin'), failing to describe the full experiment shown in the line plot.
- [**science**] Figure 5(b) lacks error bars on the line plot data points, yet the caption implies a comparison of success rates which typically requires uncertainty quantification.
- [**writing**] Figure 6: The caption for panel (c) is truncated ('(c) T2A [t2a_combined.pdf]'), omitting the description of the training duration ablation shown in the plot.

- [**science**] Figure 6: Panel (b) heatmap labels 'SFT: Beta' and 'SFT: Sig-Norm' are swapped relative to the caption's claim that 'Sigmoid-Normal at T2A and Beta at SFT' is optimal; the visual data contradicts the text description.
- [**science**] Figure 7: The caption describes the top-right panels as showing a 'clean up the table' task with sequential pick-and-place into a bin (blue umbrella, toy duck, bottled yogurt), but the images show the robot grasping a green broccoli and a pink bottle, and the bin is empty in the final frame, contradicting the claim of successful sequential placement.
- [**writing**] Figure 7: The caption lists 'blue umbrella' as an object in the top-right task, but the object being manipulated is clearly a blue plush toy or stuffed animal, not an umbrella.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and architectural shorthand that are not defined at their first occurrence, creating a barrier for non-specialist readers.

In Section 2.2, the terms "DiT" and "AdaLN" are used immediately without expansion. While standard in recent diffusion literature, a general reader needs to know these stand for "Diffusion Transformer" and "Adaptive Layer Normalization." Similarly, in Section 3.1, the training stages are referred to as "T2A" and "CPT" in the text and figure captions, but the acronyms are never explicitly defined as "Text-to-action" and "Continued pretraining" in a way that allows for easy reference later.

The Abstract introduces "OSR" and "OOD" without definition. "OSR" (Oracle Success Rate) is a specific metric distinct from standard Success Rate, and "OOD" (Out-of-Distribution) is a critical concept that should be spelled out. In Section 4.2, the text mentions converting an ODE to an "SDE" and using "GAE" for PPO; both "Stochastic Differential Equation" and "Generalized Advantage Estimation" should be defined.

Furthermore, Section 3.1.1 uses "SE(3)" to describe motion, and Table 3 lists "nDTW" without expansion. These terms, while standard in robotics and navigation, are jargon that excludes readers from other AI subfields. The paper would benefit from a brief glossary or consistent expansion of these terms upon first use to improve accessibility.

Action items.

- [**writing**] Define 'DiT' (Diffusion Transformer) and 'AdaLN' (Adaptive Layer Normalization) at first use in Section 2.2. Currently, these acronyms appear without expansion, excluding readers unfamiliar with specific diffusion architecture variants.
- [**writing**] Define 'T2A' (Text-to-action) and 'CPT' (Continued pretraining) before using them as standalone labels in Section 3.1 and Figure 2. The text introduces them as bolded phrases but does not explicitly state the acronym mapping for future reference.

- **[writing]** Define 'OSR' (Oracle Success Rate) and 'SR' (Success Rate) in the Abstract and Section 5.1.1. While 'SR' is common, 'OSR' is a specific metric that should be spelled out upon first mention to ensure clarity for non-specialists.
- **[writing]** Define 'OOD' (Out-of-Distribution) at its first appearance in the Abstract. The term is used immediately to describe generalization results without explanation, which may confuse readers from adjacent fields.
- **[writing]** Define 'SDE' (Stochastic Differential Equation) in Section 4.2 when discussing the conversion of the flow-matching ODE. The text assumes the reader knows the mathematical transformation without defining the acronym.
- **[writing]** Define 'GAE' (Generalized Advantage Estimation) in Section 4.2. While PPO is well-known, GAE is a specific component of the algorithm that should be defined for a general audience.
- **[writing]** Define 'SE(3)' in Section 3.1.1 when describing wrist motion. While standard in robotics, it is a mathematical group notation that should be briefly explained (e.g., 'SE(3) rigid body transformations') for broader accessibility.
- **[writing]** Define 'nDTW' (normalized Dynamic Time Warping) in Table 3. The metric is listed in the header without expansion, making it opaque to readers not specializing in navigation evaluation metrics.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a unified VLA model with a clear four-stage training recipe and extensive empirical results. However, there are specific logical gaps between the claims made and the evidence provided, particularly regarding the scope of "trajectory prediction" and the justification for architectural ablations.

First, the Introduction and Abstract claim that Qwen-VLA unifies "manipulation, navigation, and trajectory prediction." While the model is evaluated on manipulation (LIBERO, RoboTwin) and navigation (R2R, RxR), the navigation evaluation relies entirely on standard VLN metrics (Success Rate, Oracle Success, SPL). There is no evaluation of continuous trajectory prediction or motion forecasting (e.g., on datasets like nuScenes or Waymo, despite the mention of driving data in the pretraining mixture). The inclusion of "trajectory prediction" in the problem formulation suggests a capability that is not validated in the Experiments section, creating a disconnect between the proposed scope and the demonstrated results.

Second, the justification for excluding explicit state conditioning (Section 5.2.4) appears logically inconsistent with the reported ablation data. The authors state that state conditioning yields "marginal gains" and is omitted to reduce complexity. However, Table 5 shows that state conditioning improves RoboTwin-Hard by 1.3 percentage points (87.4% to 88.7%). In contrast, Table 4 shows that the RL post-training stage only improves SimplerEnv-OOD by 0.4 percentage points (31.6% to 32.0%). If a 0.4pp gain from RL is considered significant enough to include in the final "Instruct" model, the logical basis for discarding a 1.3pp gain from state conditioning is weak, unless a specific cost-benefit analysis (e.g., inference latency or data collection difficulty) is provided, which is absent.

Finally, the claim of "76.9% average OOD success" on real-world ALOHA (Abstract, Section 5.1.2) relies on Table 2, which aggregates performance across five categories: Color, Instance, Position, Background, and Instruction. The paper does not explicitly define the "Position" OOD category (e.g., is it object position or robot start position?) nor does it detail the specific distribution shifts used for the other categories. Without a clear definition of the OOD protocol, the causal link between the "embodiment-aware prompt conditioning" and the reported generalization performance remains ambiguous. The high performance could be attributed to the specific nature of the test set rather than the proposed method.

Action items.

- [**science**] The claim that Qwen-VLA unifies 'trajectory prediction' (Intro) is unsupported. Navigation results (Table 3) report standard VLN metrics (OS, SR) but do not evaluate continuous trajectory prediction or motion forecasting, creating a gap between the stated scope and validation.
- [**writing**] The abstract claims 76.9% OOD success on ALOHA, but Table 2 lists 'Position' as an OOD category without defining if this refers to object placement or robot start position. The specific OOD protocol is undefined, making the causal link between the training method and reported generalization ambiguous.
- [**science**] The ablation in Section 5.2.4 discards state conditioning as 'marginal' despite a +1.3pp gain on RoboTwin-Hard. This contradicts the inclusion of RL, which only yields +0.4pp on SimplerOOD (Table 4). The logical justification for discarding a larger effect size is inconsistent.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the provided empirical evidence, specifically regarding the scope of unified capabilities and the interpretation of real-world results.

First, the Abstract and Introduction explicitly state that Qwen-VLA unifies "manipulation, navigation, and trajectory prediction" and supports "driving/motion forecasting" (Section 2.1). The authors cite autonomous driving VQA data (nuScenes, Waymo) as part of the pretraining mixture (Section 4.1.4). However, the Experiments section (Section 5) contains **no quantitative evaluation** on autonomous driving or trajectory prediction benchmarks. The model is only evaluated on manipulation (LIBERO, ALOHA, etc.) and navigation (R2R, RxR). Claiming the model unifies these capabilities without demonstrating performance on the driving/trajectory tasks constitutes a significant overreach. The results only support a unified model for manipulation and navigation, not driving.

Second, the Abstract states the model attains "76.9% average OOD success in real-world ALOHA experiments." This phrasing is misleading. Table 3 (OOD performance) shows a 76.9% average *specifically* for the OOD generalization categories (Color, Instance, Position, etc.). Table 2 (In-domain performance) shows an 83.6% average for standard tasks. By presenting the OOD figure as the primary "real-world" metric in the abstract, the paper obscures the fact that the model performs even better on in-domain tasks, and the "76.9%" figure is not a general real-world average but a specific subset. This is a minor writing issue but affects the accuracy of the high-level summary.

Third, the Conclusion and Abstract claim the model achieves "strong generalist performance" across "egocentric action modeling." While 6.0% of the pretraining data is egocentric (Section 4.1.1), there are no specific benchmarks or results reported for egocentric action tasks (e.g., Ego4D, EPIC-KITCHENS action prediction). The model is only evaluated on third-person robot manipulation and navigation. The claim of unified egocentric action modeling is not supported by the experimental section.

Finally, the claim that the model "unifies... trajectory prediction" (Section 1) is unsupported by any trajectory prediction metrics (e.g., ADE/FDE on driving datasets). The "Unified Action and Trajectory Representation" section (2.4) describes the *capability* to represent trajectories, but without evaluation on a trajectory prediction task, the claim of unification in this domain is speculative.

The paper should either remove the claims regarding driving/trajectory prediction and egocentric action modeling from the Abstract and Introduction, or include the corresponding experimental results to substantiate these claims.

Action items.

- [**science**] The abstract and introduction claim Qwen-VLA unifies 'trajectory prediction' and 'driving' (Sec 1, Sec 2.1), yet the experimental section (Sec 5) provides no quantitative results on autonomous driving benchmarks (e.g., nuScenes, Waymo) despite listing driving VQA data in the pretraining mixture. This overstates the model's demonstrated capabilities.
- [**writing**] The claim of '76.9% average OOD success in real-world ALOHA experiments' (Abstract) conflates in-domain and out-of-distribution results. Table 3 shows 83.6% in-domain and 76.9% OOD. The abstract phrasing implies the 76.9% figure applies to the general real-world evaluation, obscuring the distinction between standard and OOD performance.
- [**science**] The paper claims 'strong generalist performance' across 'navigation, manipulation, and egocentric action modeling' (Abstract, Conclusion). However, the navigation results (Table 4) show Qwen-VLA-Instruct trailing StreamVLN in Success Rate (57.5% vs 56.9% is a marginal lead, but SPL is lower) and the egocentric data is only used for pretraining

without specific egocentric action benchmarks reported. The 'unified' claim overreaches the specific benchmark evidence.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a unified Vision-Language-Action (VLA) model with significant potential for embodied intelligence. However, from a safety and ethics perspective, the paper currently lacks necessary disclosures regarding data provenance, privacy, and potential dual-use risks.

Data Privacy and Consent: The authors state in Section 4.1.1 (Egocentric Human Data) that they incorporate 6.0% of data from Ego4D, EPIC-KITCHENS, EgoDex, EgoVerse, and Xperience. While these are public datasets, the paper does not include a specific section or paragraph addressing the ethical handling of human data. The review requires an explicit statement confirming that the original datasets were collected with appropriate informed consent, that Personally Identifiable Information (PII) was adequately anonymized, and that the specific usage in this VLA training pipeline complies with the original data licenses (e.g., non-commercial vs. research use). Without this, the ethical validity of the training data is unclear.

Dual-Use and Physical Safety: The model is evaluated on navigation tasks (Section 5.1.3) and incorporates autonomous driving data (Section 4.1.4). The paper claims the model can generate actions for "navigation" and "trajectory prediction." There is no discussion of the safety implications of deploying such a generalist model in real-world, unstructured environments. Specifically, the authors should address:

1. **Dual-Use Risk:** Could this model be repurposed for surveillance or autonomous weaponization?
2. **Physical Safety:** The RL stage (Section 3.2) uses sparse binary rewards in simulation. There is no mention of safety constraints (e.g., collision avoidance, force limits) being explicitly enforced during the RL optimization. If the model learns to maximize success at the cost of safety (e.g., crashing into objects or humans to achieve a goal), this is a critical safety gap. A "Safety and Limitations" section must be added to discuss these risks and propose mitigation strategies (e.g., safety filters, human-in-the-loop constraints) for real-world deployment.

Recommendation: The paper should be revised to include a dedicated "Ethical Considerations" or "Safety and Ethics" section. This section must detail the consent and privacy status of the human-centric datasets used and provide a frank discussion of the potential for physical harm and dual-use risks associated with a generalist robot controller, along with proposed safeguards.

Action items.

- [**writing**] The paper incorporates 6.0% egocentric human data (Ego4D, EPIC-KITCHENS, etc.) but lacks a dedicated 'Ethical Considerations' or 'Data Privacy' section. Explicitly state the consent status of these datasets, how PII was handled, and confirm compliance with the original data licenses regarding commercial or research use.

- [**writing**] The model is trained on autonomous driving VQA data (nuScenes, Waymo) and deployed for navigation. A 'Dual-Use' or 'Safety' discussion is missing. Address potential risks of deploying this generalist model in unstructured real-world environments (e.g., unintended physical harm, navigation failures in public spaces) and propose mitigation strategies or deployment constraints.
- [**writing**] The RL stage uses sparse binary rewards in SimplerEnv. Clarify if any safety constraints (e.g., collision avoidance, force limits) were explicitly enforced during the RL rollout or if the model relies solely on the pre-trained policy's implicit safety. If safety constraints were not explicitly modeled, acknowledge this as a limitation for real-world deployment.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a unified Vision-Language-Action (VLA) model with ambitious claims regarding generalization across tasks, environments, and embodiments. While the scale of the pretraining data and the architectural design (DiT-based flow matching) are impressive, the scientific evidence supporting the central claims of robustness and superiority lacks necessary statistical rigor.

The primary concern is the absence of variance metrics. Tables 1, 2, and 3 report single-point success rates (e.g., 97.9% on LIBERO, 83.6% on ALOHA) without standard deviations, confidence intervals, or p-values. In robotics and embodied AI, performance can vary significantly based on random seeds, initial object poses, and environmental noise. Without reporting the standard deviation over multiple seeds (typically N=5 or N=10) or trials, it is impossible to determine if the reported improvements over baselines (e.g., the +35.4pp gain over $\pi_{0.5}$ in OOD tasks) are statistically significant or potentially due to favorable random initialization.

Furthermore, the real-world evaluation on ALOHA (Table 2) lacks transparency regarding the experimental protocol. The manuscript states the model was evaluated on "short-horizon and long-horizon task categories" but does not specify the number of trials per task, the number of seeds, or the specific randomization of object placements used during evaluation. Given the high success rates reported (e.g., 98.7% for Bowl Stacking), the lack of error bars raises concerns about the robustness of these results to minor perturbations.

Finally, the Out-of-Distribution (OOD) claims on the DOMINO benchmark (Table 4) rely on a zero-shot setting. While the result of 26.6% is notable, the paper does not detail the distribution of the 35 suites or the number of episodes per suite used to calculate this average. If the evaluation was conducted on a small number of episodes per suite, the result could be highly volatile. The authors must provide the sample size (N) and variance for all key quantitative results to substantiate the claim of "robust multi-task performance and OOD generalization."

Action items.

- [**science**] Report statistical significance (e.g., p-values, confidence intervals, or standard deviations) for the reported success rates in Tables 1, 2, and 3. The current presentation of single-point estimates (e.g., 97.9%, 83.6%) without variance metrics makes it impossible to assess the robustness of the gains over baselines like pi_0.5 or GR00T.
- [**science**] Clarify the sample size (N) and number of random seeds used for the real-world ALOHA experiments (Table 2). The text mentions 'real-world ALOHA experiments' but does not specify if the reported 83.6% average is an aggregate of multiple trials, seeds, or a single run, which is critical for evaluating the reliability of the OOD claims.
- [**science**] Provide a detailed description of the randomization strategy and sample size for the DOMINO dynamic manipulation benchmark (Table 4). The claim of 26.6% zero-shot success is significant; the review requires evidence that this result is not an artifact of a specific seed or a small, non-representative subset of the 35 suites mentioned.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the manuscript is insufficient to support the robustness of the claimed improvements. While the paper presents extensive numerical results across multiple benchmarks (LIBERO, Simpler, RoboTwin, DOMINO, VLN-CE), it consistently reports single-point success rates (e.g., 97.9%, 73.7%) without accompanying measures of variance such as standard deviations, confidence intervals, or standard errors. This is a critical omission in empirical robotics and machine learning research, where performance can vary significantly based on random seeds, environment initialization, or stochastic policy execution.

Specifically, in Tables 1 through 10, the authors claim statistically significant improvements (e.g., "outperforming $\pi_{0.5}$ by 35.4 percentage points" in Table 3, or "+2.9pp" gain from RL in Table 12). Without reporting the variance across independent trials or seeds, it is impossible to determine if these differences are statistically significant or merely artifacts of random variation. The text mentions using "128 parallel envs" for RL rollouts (Section 5.2), but this refers to parallelization within a single run, not the number of independent experimental seeds required to estimate population variance.

Furthermore, the ablation studies (Tables 11-14) involve testing multiple configurations (e.g., different projection heads, state conditioning strategies, T2A hyperparameters). The manuscript does not address the multiple-comparisons problem. When testing many hypotheses, the probability of finding a "significant" result by chance increases. The authors should either apply a correction method (e.g., Bonferroni, False Discovery Rate) or explicitly frame these as exploratory analyses where the lack of correction is acknowledged.

Finally, the sample sizes for the real-world ALOHA experiments (Table 2) are not explicitly stated. How many trials were conducted per task category? Without this information, the reliability of the 83.6% average success rate cannot be assessed. The authors must provide the number of seeds/trials, report variance metrics (mean ± std), and clarify the statistical testing methodology used to validate their claims.

Action items.

- [**science**] Report confidence intervals or standard deviations for all success rates (Tables 1-10). Single-point estimates prevent assessing statistical significance of claimed gains (e.g., +2.9pp RL gain).
- [**science**] Clarify the number of independent seeds or trials used for aggregate success rates. The text mentions '128 parallel envs' but does not specify if benchmark scores are averages over multiple seeds, which is critical for variance estimation.
- [**science**] Address the multiple-comparisons problem in ablation studies (Tables 11-14). With many configurations tested, lack of correction (e.g., Bonferroni) risks inflating Type I error rates for claimed 'marginal gains'.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, but the writing quality requires minor revisions to ensure maximum clarity and readability for a broad audience. The prose is generally fluent, yet several specific areas suffer from undefined terminology, inconsistent notation, and ambiguous phrasing that could hinder comprehension.

First, in Section 3.2 under "Model Architecture," the authors mention the use of "multi-section RoPE." This is a specific technical implementation detail that is not standard terminology in the general VLA literature. Without a brief definition or a citation to the specific method (e.g., YaRN, LongRoPE, or a specific Qwen implementation note), readers may be confused about how the positional embeddings are handled. This should be clarified to ensure the architectural contribution is fully understood.

Second, the notation used for the real-world model variants in Section 5.1.2 is problematic. The LaTeX code `${\\text{Qwen-VLA-aloha}}_{\\text{w/ pretrain}}$` results in a visually cluttered and non-standard mathematical notation when rendered in the text. Phrases like "Qwen-VLA-aloha w/ pretrain" are difficult to read and break the flow of the narrative. It is strongly recommended to adopt a cleaner naming convention, such as “Qwen-VLA-Aloha-FT” or “Qwen-VLA-Aloha (Pretrained),” which would improve readability in both the text and the tables (e.g., Table 3 and Table 4).

Third, the abstract presents a series of high-precision metrics (e.g., “97.9% on LIBERO,” “73.7% on Simpler-WidowX”) without explicitly contextualizing them as absolute success rates or relative improvements. While the body of the paper likely clarifies this, the abstract should be self-contained. Ambiguity here could lead readers to misinterpret the magnitude of the results. A brief phrase such as “achieving an absolute success rate of 97.9%...” would resolve this.

Finally, there are minor instances of passive voice and slightly dense sentence structures in the “Limitations and Future Work” section that could be streamlined for better impact. For example, the sentence “Joint training across vision-language understanding, navigation, and action generation introduces optimization trade-offs” is grammatically correct but could be more direct.

Overall, the paper is well-written, but addressing these specific points will significantly enhance its clarity and professional presentation.

Action items.

- [**writing**] In Section 3.2 (Model Architecture), the phrase ‘multi-section RoPE’ is used without definition. Define this term or cite the specific method (e.g., YaRN, LongRoPE) to ensure clarity for readers unfamiliar with this specific implementation detail.
- [**writing**] Section 5.1.2 (Real World Manipulation) introduces the model variant notation Qwen-VLA-aloha$_{\text{w/ pretrain}}$. This notation is visually cluttered and inconsistent with standard academic style. Consider renaming the model to ‘Qwen-VLA-Aloha-FT’ or similar for better readability in text and tables.
- [**writing**] The abstract lists specific performance metrics (e.g., ‘97.9% on LIBERO’) but does not explicitly state the baseline or comparison metric (e.g., ‘improvement over SOTA’ or ‘absolute score’). Clarify whether these are absolute success rates or relative improvements to avoid ambiguity.